The Effect of Outlier Treatment on Predictive Modeling Performance: A Comparative Study

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Abstract

Outliers are a common problem in real-time datasets. They may arise because of errors in measurement or unusual events. The presence of outliers crucially impacts the statistical process and machine learning analysis, as they provide biased estimates and reduce the performance of the prediction model. There are several techniques that help in the treatment of outliers, such as removal of outliers, winsorization, and transformation; they aim to improve model performance by treating the outliers using Python and R. These techniques limit the influence of outliers and produce effective analysis and results. This study mainly focuses on the effect of treating outliers and the effects of not treating the outliers on a model based on their metrics. Using the weather dataset from the Hyderabad region, we applied different models to check the performance with and without outlier treatment. As such we are motivated to set our objectives to examine the effect of outlier treatment on predictive model with the help of R^2 metric. The second objective is to, determine how a model performs across different outlier handling approaches. The paper concludes with practical guidelines for researchers and practitioners on when and how outlier treatment should be applied.

Keywords: Outlier Treatment, R^2, Trimming, Winsorization, Machine Learning

1 Introduction

Now-a-days the world is completely data-driven across all the domains such as health-care, finance, weather, agriculture. These domains are gradually depending on machine learning and statistical processes to analyze the data and get the insights for effective decision making. The robustness of these models depends on the quality of the data used for analysis. These real-time datasets are rarely pure and they often contain outliers, missing values, inconsistent data points. Among the difficulties, outliers, the data points which are significantly far from the remaining pattern, are the most encountered issues among the datasets.

Outliers may arise because of several reasons. During data collection, the outliers can be caused by manual typing errors or improper responses. When data collected from measurements systems, the data from faulty sensors, improper apparatus gives rise to the values outside the specific range. Whereas the data collection from real-world may produce extreme values due to rare and genuine processes such as heavy rainfalls, pandemics, and failure or uprise of stock market. Whatever may be reason for the arising of outliers, they influence the data modelling and prediction processes heavily if not taken care.

Most of the statistical processes and machine learning methods assume that data follows a specific distribution, near to normal distribution, which can become difficult with the presence of outliers. These outliers impact heavily on the descriptive statistic measures like mean, standard deviation, skewness and also influence the coefficients of the models. Due to these concerns, the question of handling outlier had become the major problem in all fields which practice machine learning.

Outliers can be broadly categorized into three types:

- Global outliers: The data points which fall outside the distribution range.
- Contextual outliers: The data points that rarely occurred with a specific context.
- Collective outliers: The data points that collectively stay away from the remaining data points.

Primary and easy way of dealing with outliers is removing them using z-score or IQR method. Removing outliers may also lead to bias. Removal of outliers is also known as Trimming. Capping, also known as winsorization, replaces the extreme values with boundary values. Another solution for treating them can be transforming the values using log transformation, box-cox transformation.

While the importance of handling outliers is well identified. Most existing literature focuses on outlier handling in pre-processing step without noticing the impact on model performance. Most of the users remove outliers immediately in pre-processing step, without thinking whether the removal of outliers does any impact on the model performance or not.

This research addresses the gap by performing the study in a comparative way with the use of predicting models trained on the data with and without the treatment of outliers. We use a weather dataset from the Hyderabad region, in which the data specifically contains measurement errors such as recording temperature, humidity, and also the unusual and genuine errors which are caused by heavy rainfalls, cyclones. This data also helps us in the role of outliers with and without treatment.

2 Literature Review

The research has developed a method called Double Weighting and LOF to improve the detection of the abnormal situations in furnaces. This method enriches checking indicators and stabilize the results. The tests performed on the real furnace data presents that it performs better than the traditional methods [1] . The study had given a brief review of the abnormal observation detection methods for time series data in the field of wireless sensor networks, including both outlier detection and abnormal pattern frameworks. Their research reveals six major outlier detections and the limitations and also the guidance for handling the sensor-based data abnormalities [2]. The research initiated an outlier detection method which combines Principal Component Analysis and K – means clustering for isolating abnormal observations. Their research contrasts the above

hybrid model with the traditional methods such as standard deviation and box plot. The research results the superiority of Principal Component Analysis based method [3]. The researchers worked on a method for detection and estimation of both position and value of the outliers in time series which is based on maximum likelihood function. The results shown better performance when applied to Japan's industrial production index dataset [4]. The study had given an outlier detection model for time series which is based on clustering, it is capable of non-linear trends to efficiently identify the outliers without prior labels. The study results, include testing on real dataset of insurance and also improve the decision making by detecting the abnormal patterns [5]. The study had proposed a method for detecting outliers in high dimensional metabolomics datasets. The research results that the method improves the performance and accuracy of clustering while compared to the traditional methods when data is clean [6]. The research had given GHOA, which refers to global high dimensional outlier detection model, that categorizes data before analysis to improve the performance of finding the outliers. The results of the study show that GHOA performs better than the techniques which are existing [7].

The study presented a comparison study of three major outlier detection methods such as cluster – based, density – based and distance – based to identify the outliers among different data types. Their study results the strengths and the effectiveness of each method, resulting to give the best – performing method for efficient outlier detection [8]. The research proposed a method ODCA, which is a cross correlation dependent outlier detection method, used to handle the higher dimensional time series data. Their study analysis among several datasets proved that OCDA is better than other detection methods in accuracy and computational efficiency [9]. The study demonstrated the limitations of traditional methods in outlier detection for complex data types and also give importance to entropy and spectral based methods. The research also provided needs in future for enriching scalability and performance [10]. The research initiated SODEP, which is a subspace outlier detection method based on ensemble, which holds several PCA based subspaces to overcome the problem of high dimensionality. Their study results that the aggregate of outlier scores across the relevant sub spaces gives efficient

performance in detection process compared to traditional techniques [11]. The study implemented a PCA based K - means on high dimensioned smart meter data and clustering and distance - based thresholding method detects accurate and lower complexity outliers on data [12]. The research introduced a method for outlier detection for data involving non-Gaussian based on normalizations and transformations. Their study reveals that multivariate Johnson transformation method gives better performing results for outlier detection [13]. The research introduced ClaCO, which is a outlier detection model based on social network analysis, that transforms tabular labelled data into non linear data structures called graphs and evaluate the node's consistency using CSoN. The results prove that removal nodes which are low consistent enriches the performance among multiple real time datasets [14].

The study focused on a comparison based on supervised and unsupervised models for the identification of types of outliers such as global, contextual and collective. Their research features the difference in the methods when executed on standard and real-time datasets. The research reveals how the performance of model changes based on characteristics of data and outliers [15].The research introduced an outlier detection method which is based on clustering, that identifies a set of samples using hierarchical clustering and gets a statistical threshold to check for outliers. The results on IRIS data shows that the method is effective [16]. The study demonstrated that transformation method called Hilbert – Huang transformation is effective compared to the wavelet techniques to detect the outliers in non-linear data. The enrich HHT with EEMD to get more efficient analysis results [17]. The study introduced a outlier filtering method based on analyzing neighbor distance distributions and removing the observations whose mean distances deviate from the Gaussian thresholds. Their study helps in removing the sparse measurement errors in the point cloud registration [18]. The study demonstrated a reconstruction and outlier identification technique by combining several detection models using a LSCP framework and by using light weight gradient boosting for repairs in photovoltaic power dataset. The result evaluation shows that this method is efficient in detecting and reconstructing outliers in a cost-efficient way [19]. The research proposed a visual system called Voodds

to explain how outliers are detected in data streaming using different views like data distribution, box graph combinations, broken line hisogram and correlation plot [20].

3 Methodology

3.1 Study Area

The present research takes place with the metrological information available of the Hyderabad region; a city located in the Telangana state of Southern India. Hyderabad city is characterized by high summers, cool winter months, and a defined rainy season.
The annual average rainfall for the region is approximately 750 – 850 mm. The temperature in the city has a wide variation from 12°C during winter months to beyond 40°C during summer months making this city exposes severe changes in daily temperature leading to very unusual and rare temperature events, that may result in outliers.

3.2 Data Source

The weather data used in this study was gathered from the NASA Prediction of Worldwide Energy Resources (POWER) project, which is an open-access platform providing the satellite-derived weather datasets. The daily observations for the Hyderabad region were extracted using the parameters T2M_MAX, T2M_MIN, RH2M, PRECTOTCORR, WS2M, ALLSKY_SFC_SW_DWN from the website. The variables expand over the span of 1984-2025, covering daily weather data of the variables.

3.3 Outlier Detection using Interquartile Range (IQR) Method

The Inter Quartile Range is the measure of statistics which calculated the spread of data, by calculating the difference between third quartile (75th percentile) and the first quartile (25th quartile) of the data.

$$\text{IQR} = Q_3 - Q_1 \tag{1}$$

A data point is considered as an outlier if lies below the lower bound and above the upper bound. They are calculated using

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR} \tag{2}$$

$$\text{Upper Bound} = Q_3 + 1.5 \times \text{IQR} \tag{3}$$

If any datapoint < LowerBound and datapoint > UpperBound, then these datapoints are considered as outliers.

3.4 Outlier Treatment

We applied three strategies for treatment of outliers.

- Removal of Outliers: The outliers are removed from the entire data. This process is also known as Trimming.
- Winsorization: The detected outliers are often replaced by the boundary data points. This process is also known as capping. If the outliers lie below the LowerBound, then these outliers are replaced by the value of LowerBound. Similarly, if the outliers lie above the UpperBound, then these outliers are replaced by the value of UpperBound.
- Transformation: Transformation help to treat the outliers by transforming them into the possible values to reduce the impact.
  - Log transformation: The original data values are transformed into Logarithmic values. This method can make the larger values look smaller.

    $$x' = \ln(x) \tag{4}$$

    where x is the actual value and x' is the transformed value.
  - Box-cox transformation: It is the generalized version of log transformation.

Box-cox transformation uses a parameter (λ) to transform the values.

$$y' = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda}, & \lambda \neq 0, \\ \ln(y), & \lambda = 0. \end{cases} \tag{5}$$

where y = original value (must be $y¿0$).

λ = transformation parameter chosen to best approximate normality.

λ is basically calculated using maximum likelihood estimation (MLE).

3.5 Models

3.5.1 Multiple Linear Regression

The statistical technique used to fit the linear model between a target variable and two or more predictor variables. The representation is as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon \tag{6}$$

Where:

Y = dependent variable,

$X_1 + X_2 + \cdots + X_n$= independent variables,

$\beta_0 + \beta_1 + \beta_2 + \cdots + \beta_n$ = coefficients,

ε= error term.

3.5.2 XG Boost

It is an ensemble learning model, which aggregates the results of the different models in a sequential manner. It is known for its efficiency in handling missing values and non-linear relationships.

3.6 Coefficient of Determination

The percentage of the dependent variable's variance that can be predicted from the independent variable is known as the coefficient of determination. It indicates the goodness of fit of a model. R^2 values always lie between 0 and 1.

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} \tag{7}$$

Where:

$SS_{\text{res}} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ denotes the residual sum of squares,

$SS_{\text{tot}} = \sum_{i=1}^{n}(y_i - \bar{y})^2$ denotes the total sum of squares.

4 Results and Discussion

4.1 Data Representation

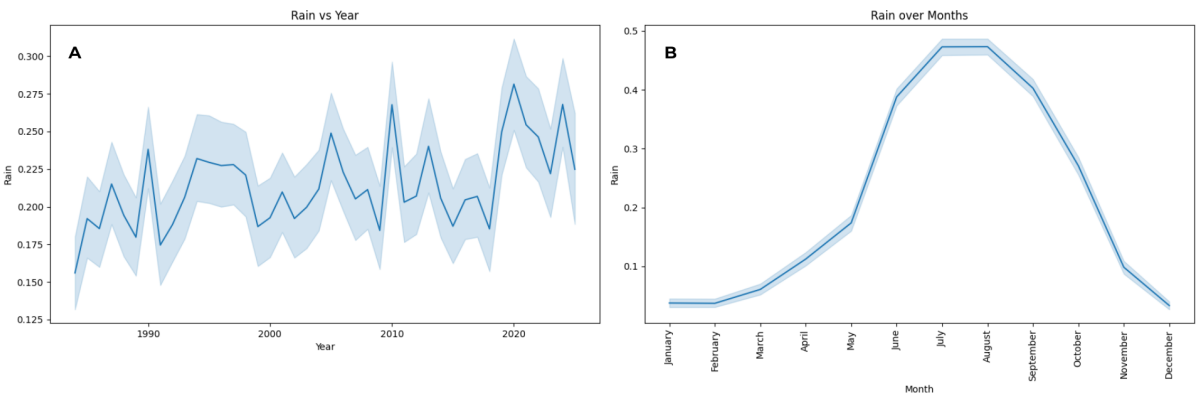

Figure 1: Rain Over Time

- **Figure - 1 (A) - Rain VS Year:** Rain shows a variation every year with no trend, showing ups and downs due to weather conditions.
- **Figure - 1 (B) - Rain VS Month:** A specific rainy season can be observed during July - August, indicating rainy season and decreasing after September.

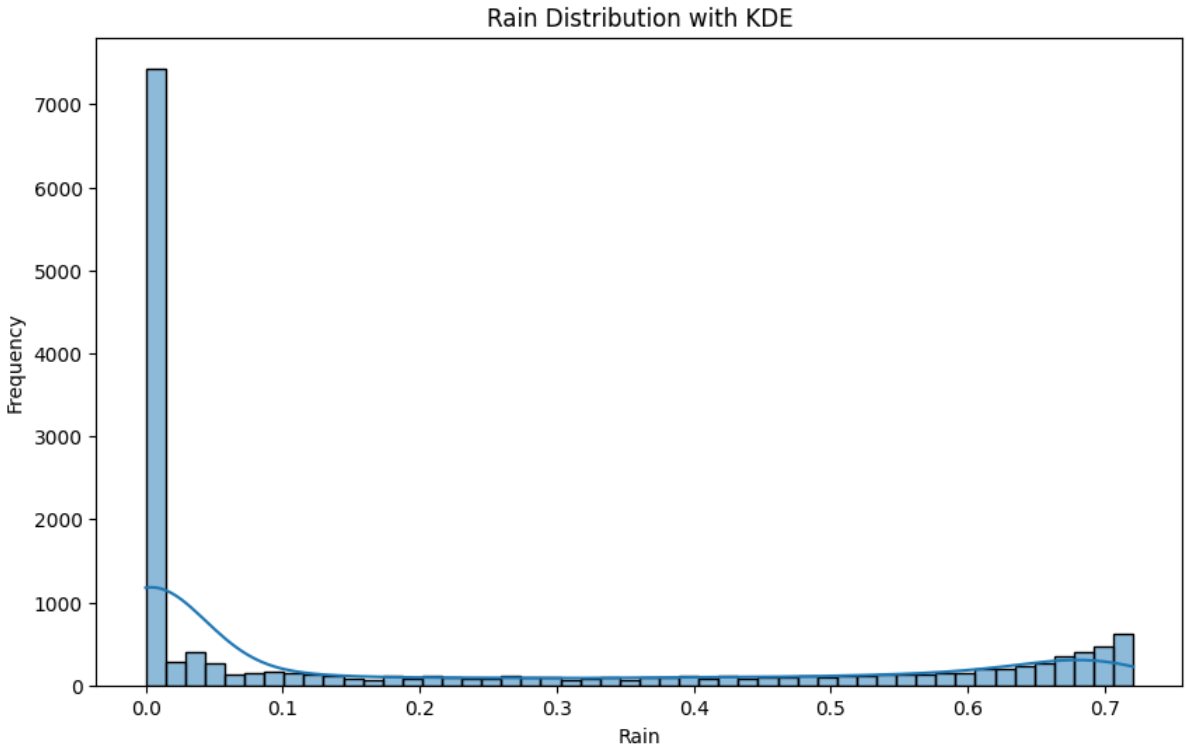

Figure 2: Distribustion of Rain

In Figure - 2, the distribution of the Rain variable is mostly right - skewed, with almost many observations around zero and also indicating heavy raining events occasionally near the tail.

4.2 Outlier Detection

Below are the box-plots used for the detection of outliers in the data and after using the outlier treatments on the data.

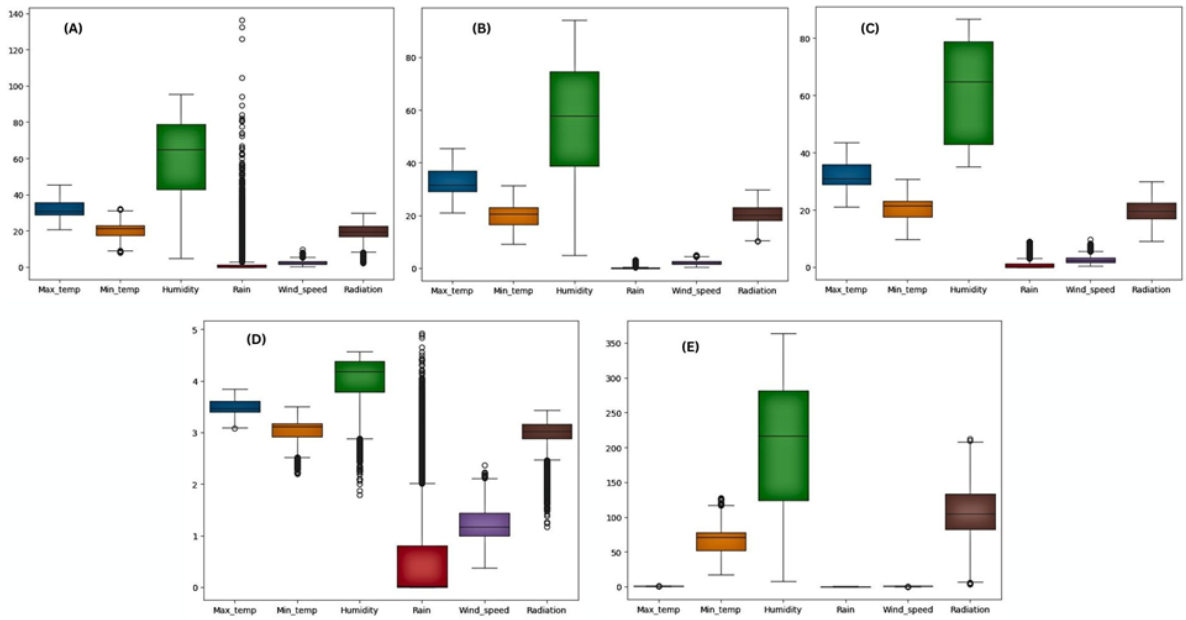

Figure 3: Implementation of Various Outliers Detection Techniques

- **Figure - 3(A) - Original Data:** This box plots depicts the original dataset's distribution. It shows the presence of outliers, specifically in rainfall and wind speed.
- **Figure - 3(B) – Trimming:** After trimming, the outliers which are extreme are removed, which results in more proper distributions.
- **Figure - 3(C) – Winsorization:** The extreme values are replaced by either upper bounds or lower bounds. This preserves the values and does not affect the size of the data. The distribution is balanced.
- **Figure - 3(D) – Log Transformation:** The distributions are skewed, especially in rainfall and radiation. But show more variability.
- **Figure - 3(E) – Box – Cox Transformation:** This Box–Cox transformation generally normalizes the data effectively by reducing skewness and stabilizing variance

The results of the outlier treatments are evaluated by the machine learning models i.e., Linear Regression and XGBoost.

4.3 Outlier Treatment and Evaluation

The treatment of outliers plays an important role in the fitting of predictive models and their performance. R^2 explains the variance among the data, if the value is higher the model fits the best, enabling direct comparisons of the models. It always lies between 0 to 1.

4.3.1 Linear Regression

Table 1: R^2 Values of the different approaches using Linear Regression

Approach	R^2 **Value**
With Outliers	0.5783
Trimming	0.5308
Winsorization	0.5900
Log Transformation	0.5363
Box–Cox Transformation	0.5900

While using linear regression in the Table 1, the table depicts the base R^2 value as 0.5783. After the use of different outlier treatment techniques, there is a decrease in the performance of model during the use of trimming and log transformation, meanwhile winsorization and Box - Cox Transformation (i.e., with R^2 of 0.59). The study suggests that the winsorization and Box - Cox transformation performs more effectively with Linear Regression and they also preserve the extremes values by replacing with upper and lower bounds and transforming respectively by reducing the effect of outliers. The prediction equations are as follows:

$$\begin{aligned} \text{Rain} = 0.74\,\text{Max_temp} - 0.18\,\text{Min_temp} - 0.29\,\text{Radiation} \\ + 0.019\,\text{Humidity} - 0.49\,\text{Wind_speed} - 0.417007 \end{aligned} \tag{8}$$

$$\begin{aligned} \text{Rain} = 61160\,\text{Max_temp} + 0.1271\,\text{Min_temp} - 33.19\,\text{Radiation} \\ - 0.055\,\text{Humidity} - 29.79\,\text{Wind_speed} - 35010 \end{aligned} \tag{9}$$

4.3.2 XG Boost

Table 2: R^2 values of different outlier treatment approaches using XGBoost

Approach	R^2 **Value**
With Outliers	0.7497
Trimming	0.7237
Winsorization	0.7475
Log Transformation	0.7563
Box–Cox Transformation	0.7534

In Table 2, it shows the results of XGBoost model and the effect of different outlier treatments. The base R^2 value for the model is 0.7497. The outlier treatment is stable and less deviating compared to linear regression. After the treatment of outliers, while trimming reduces the value slightly and log transformation and Box-Cox transformation improved the result to some extent. In XGBoost, the values range from 0.7237 to 0.7563, because XGBoost model is a tree-based model, which uses the ensemble learning method for the fitting. It is less sensitive to the outliers and provide better results compared to the linear regression.

The overall results highlight that the choice of outlier treatment depends on the model being used. Linear regression is more beneficial from the treatments like winsorization and Box-Cox transformation, whereas XGBoost is consistent even with outliers and shows some gain from log-based transformation. The study only shows the importance of outlier treatment during the pre-processing step.

5 Conclusion

Outlier treatment plays an important role in the performance of a linear regression model. Winsorization and Box–Cox transformation provided better results, whereas trimming and log transformation reduced the model's efficiency. For XGBoost, the situation is different, as it is a tree-based ensemble modelling technique. Tree-based models are

generally less sensitive to outliers and remain stable throughout the modelling process. The results of this study confirm that behavior for XGBoost, with log and Box–Cox transformations offering slight improvements.

Finally, outlier treatment should not be applied automatically without first assessing its effectiveness on the data and the modelling approach. The choice of outlier treatment must be made carefully for the models like Linear Regression, as they are extremely sensitive to assumptions of the distributions. But tree-based models like XGBoost can handle the outliers without heavy preprocessing, making them the best choice for datasets where managing outliers is particularly challenging.

In future this research can explore other outlier treatments and computational tradeoffs like speed, space, time for executing the treatment of outlier.

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